

1 Introduction

I have curated questions from multiple exams and compiled them below. Some questions are taken from HSC so you can look at answers by looking up that HSC paper. In the first section, we will look at Module 5. Feel free to complete the questions with your notes open as the questions are aimed for you to revise the content you already learnt. In the second and final section, I will introduce more of Module 6, starting with Year 11 revision from the Electricity topic before giving you HSC level Module 6 questions.

I want you to be able to solve these questions with sufficient understanding so I will give hint throughout as to where to start. If you need solutions or more hints, feel free to text me. I hope these questions will last you for 3 weeks so you will be occupied until I am back on the 11th of February.

2 Projectile and Circular motion

2.1 Question 1

Experienced target shooters say that they do not so much aim at the target as 'feel' for the right spot above the target. Michelle is an experienced target shooter who has scored a perfect bullseye on a target 300 m away. If the target was set at her eye level and the bullet left the rifle at a speed of 600ms^{-1} , how high above the bullseye was she actually aiming her rifle?

Click to show hint

2.2 Question 2 (Q24 2025 HSC)

Two satellites, A and B, are in stable circular orbits around the Earth. The radius of satellite A's orbit is three times that of satellite B's orbit. Both satellites have the same kinetic energy.

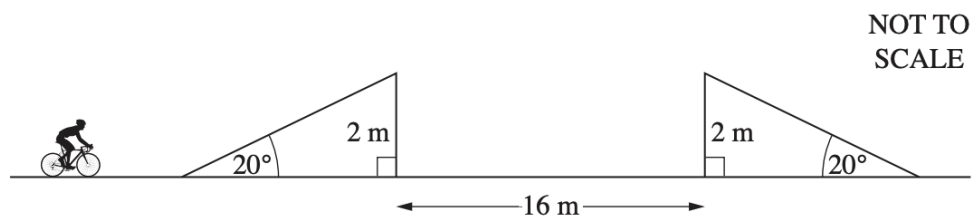
Show that the mass of A is three times the mass of B.

Click to show hint

2.3 Question 3 (Q28 2025 HSC)

Take a look at Figure 1, ignoring the labelled values for the angle of the ramp, θ , the height of the ramp, h and the length between the ramps, L , derive a general expression for the minimum speed required for the rider to land on the second ramp in terms of those variables.

A bicycle rider jumps from one ramp to a second ramp separated by 16 m as shown. The ramps are inclined at 20° and are 2 m high.



What minimum speed is required for the rider to land on the second ramp?

Figure 1: Ignore labelled values

[Click to show hint](#)

2.4 Question 4 (Q29 2025 HSC)

A mass moves around a vertical circular path of radius r , in Earth's gravitational field, without loss of mechanical energy. A string of length r maintains the circular motion of the mass.

When the mass is at its highest point B , the tension in the string is zero, shown in Figure 1

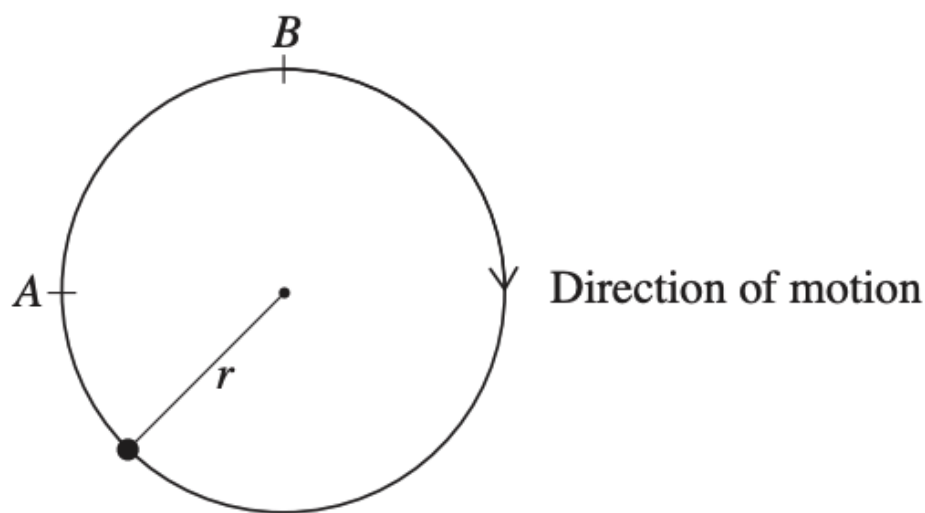


Figure 2: Trajectory of mass

- Show that the speed of the mass at the highest point, B , is given by $v = \sqrt{rg}$
- Compare the speed of the mass at point A to that at point B i.e. find k when $v_a = kv_b$

Click to show hint

2.5 Question 5 (Q25 2024 HSC)

Derive the relationship between the orbital radius of a satellite and its period,

$$\frac{r^3}{T^2} = \frac{GM}{4\pi^2} \quad (2)$$

Click to show hint

2.6 Question 6 (Q18 2023 NSB Trial paper)

Saturn has several rings, each of which consists of many small particles that orbit the planet. Saturn may be considered to be a sphere with its mass M concentrated at its centre.

- (a) Deduce that, for a particle in one ring moving in a circular orbit of radius R , the linear speed v of the particle in its orbit is given by the expression

$$GM = Rv^2 \quad (3)$$

- (b) One ring, the A ring, has an outer diameter of 2.72×10^8 m. The mass of Saturn is 5.69×10^{26} kg. A particle orbits on the outer edge of this ring. Determine the time for the particle to complete one orbit of Saturn
- (c) Explain the concept of escape speed, derive the expression for escape speed for a planet with an acceleration of free fall at its surface of g .
- (d) Calculate the escape speed for a spherical planet of radius 1.7×10^3 km having an acceleration of free fall at its surface of 1.6 ms^{-2}

Click to show hint

2.7 Question 7

A rubber stopper of mass 50.0 g is whirled in a horizontal circle on the end of a 1.50 m length of string. The time taken for time complete revolutions of the stopper is 8.00 s. The string makes an angle of 6.03° with the horizontal. Calculate:

- (a) the speed of the stopper
- (b) the centripetal acceleration of the stopper
- (c) the net force acting on the stopper
- (d) the magnitude of the tension in the string

I won't give any clues here because any hint would make the question too easy.

3 Electric fields (Revision)

3.1 Question 1

- (a) Calculate the acceleration of an electron in an uniform electric field of strength $1 \times 10^6 \text{ NC}^{-1}$
- (b) Starting from rest, how long would it take for the speed of the electron to reach 10% of the speed of light? (Ignore relativistic effects.)
- (c) What distance would the electron travel in that time?
- (d) If the answer to (c) was the actual spacing of the plates producing the electric field, what was the voltage drop or potential difference across the plates?

Click to show hint

3.2 Question 2 (Q28 2024 HSC)

Refer to Figure 3, the electron gun fires a beam of electrons at $2.0 \times 10^6 \text{ ms}^{-1}$ through a pair of parallel charged plates towards a screen that is 30 mm from the end of the plates.

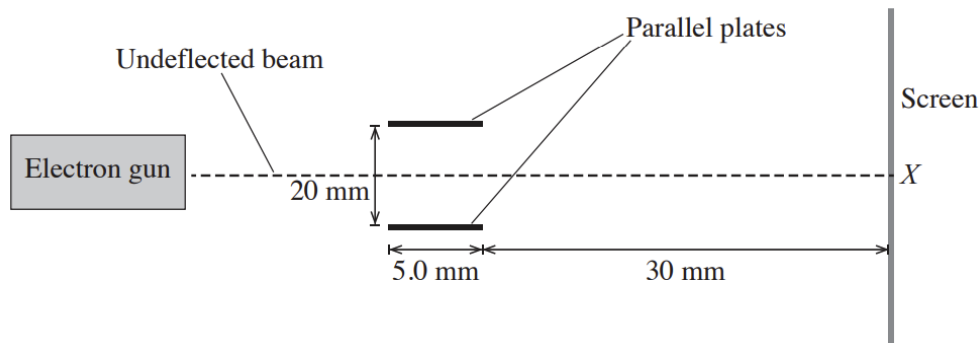


Figure 3: There is a uniform electric field between the plates of $1.5 \times 10^4 \text{NC}^{-1}$. The plates are 5.0 mm wide and 20 mm apart. The electron beam enters mid-way between the plates. X marks the spot on the screen where an undeflected beam would strike.

- Show that the acceleration of an electron between the parallel plates is $2.6 \times 10^{15} \text{ms}^{-1}$
- Show that the vertical displacement of the electron beam at the end of the parallel plates is approximately 8.1 mm
- How far from point X will the electron beam strike the screen

Click to show hint